

ISLAND FALLS, SK, Canada

WMO: 713310

Lat: 55.530N

Lon: 102.350W

Elev: 297

StdP: 97.80

Time Zone: -6.00 (W06)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.4	(c) -33.6	(d) -40.7	(e) 0.1	(f) -36.4	(g) -37.6	(h) 0.1	(i) -33.5	(j) 5.7	(k) -17.8	(l) 5.2	(m) -17.9	(n) 0.7	(o) 240	(p) 0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	27.4	17.1	25.7	16.2	24.1	15.3	18.5	25.0	17.5	23.5	16.6	22.1	2.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	11.8	21.1	15.1	11.1	20.1	14.2	10.5	19.2	53.8	25.4	50.7	23.3	47.6	22.2	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.0	(b) 5.2	(c) 4.5	(e) -40.0	(f) 30.6	(g) 2.8	(h) 1.7	(i) -42.0	(j) 31.8	(k) -43.6	(l) 32.8	(m) -45.1	(n) 33.7	(o) -47.2	(p) 34.9		
(a) 6.0	(b) 5.2	(c) 4.5	(e) -40.0	(f) 20.3	(g) 2.7	(h) 1.0	(i) -41.9	(j) 21.0	(k) -43.5	(l) 21.5	(m) -45.0	(n) 22.1	(o) -47.0	(p) 22.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.7	-19.2	-18.5	-10.2	-0.4	7.6	14.0	17.0	15.5	9.9	2.0	-8.9	-17.8
	DBStd	14.46	7.83	7.28	7.61	6.58	4.77	3.88	3.04	3.15	4.51	4.31	7.16	8.07
	HDD10.0	4495	906	797	626	312	105	11	0	2	56	249	568	862
	HDD18.3	6969	1164	1031	884	561	334	137	62	98	254	505	818	1121
	CDD10.0	605	0	0	0	2	29	129	217	172	53	2	0	0
	CDD18.3	38	0	0	0	0	1	6	21	9	2	0	0	0
	CDH23.3	502	0	0	0	0	26	98	238	112	28	0	0	0
	CDH26.7	77	0	0	0	0	3	12	46	12	4	0	0	0
Wind	WSAvg	2.0	1.9	1.8	2.0	2.2	2.3	2.0	2.0	2.1	2.1	2.2	2.0	1.8
Precipitation	PrecAvg	519	22	16	19	20	46	74	86	74	72	37	27	27
	PrecMax	669	39	38	43	42	131	204	188	138	192	66	70	53
	PrecMin	340	5	1	2	0	9	21	28	18	11	7	8	8
	PrecStd	71	9	9	11	11	26	38	36	33	43	17	16	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.6	2.1	10.6	19.6	26.3	28.2	30.2	28.2	26.3	17.6	6.8	0.2
		MCWB	-1.0	0.2	4.6	9.3	13.3	17.0	18.1	17.9	16.4	10.4	3.4	-1.2
	2%	DB	-2.5	-1.1	6.7	15.7	23.2	26.0	27.8	26.2	23.3	13.6	4.2	-2.5
		MCWB	-3.4	-2.7	1.8	7.5	12.2	15.3	17.7	17.3	15.2	8.5	1.9	-3.5
	5%	DB	-5.4	-4.1	3.6	12.4	20.8	24.1	26.0	24.4	20.4	10.9	2.0	-4.7
		MCWB	-6.1	-5.5	0.3	5.6	11.1	14.6	16.6	16.3	13.8	7.6	0.5	-5.4
	10%	DB	-7.6	-6.9	1.1	9.8	17.7	22.2	24.2	22.3	17.6	8.6	0.2	-6.3
		MCWB	-8.3	-7.9	-1.2	3.9	9.5	13.8	16.0	15.3	12.6	6.0	-1.0	-6.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.9	0.3	4.6	10.0	14.6	18.8	19.8	19.5	17.7	11.6	3.9	-0.9
		MCDB	0.3	1.9	10.2	18.5	23.8	25.3	26.3	26.1	24.6	15.5	5.9	0.3
	2%	WB	-3.4	-2.8	2.2	7.7	12.9	17.3	18.7	18.1	15.8	9.2	2.0	-3.4
		MCDB	-2.5	-1.3	6.0	15.2	21.2	23.3	25.5	24.1	21.9	12.8	3.7	-2.6
	5%	WB	-6.0	-5.4	0.5	5.9	11.5	16.1	17.9	17.0	14.5	7.7	0.5	-5.4
		MCDB	-5.4	-4.2	3.2	11.9	19.4	21.3	23.9	22.5	19.7	10.8	2.0	-4.7
	10%	WB	-8.2	-8.0	-1.1	4.2	10.1	14.9	16.8	16.1	12.8	6.1	-1.1	-6.9
		MCDB	-7.6	-6.9	1.1	9.1	17.5	20.2	22.6	21.0	17.2	8.5	0.2	-6.3
Mean Daily Temperature Range	5% DB	MDBR	9.1	12.0	13.3	12.9	13.1	12.0	11.4	10.4	10.0	6.8	6.9	8.0
		MCDBR	10.0	12.7	13.9	16.4	19.2	15.7	14.8	14.5	14.5	11.5	8.3	8.1
		MCWBR	9.4	11.1	9.9	9.5	9.6	7.1	6.8	7.0	7.6	7.3	6.3	7.6
	5% WB	MCDBR	9.9	12.3	12.4	15.6	16.8	12.1	12.6	12.1	13.2	10.3	7.7	8.4
		MCWBR	9.3	10.9	8.9	9.3	8.6	6.1	6.5	6.6	7.3	6.7	6.0	7.9
Clear-Sky Solar Irradiance	taub	0.232	0.249	0.268	0.319	0.330	0.367	0.393	0.359	0.319	0.287	0.242	0.216	
	taud	2.283	2.316	2.332	2.394	2.487	2.409	2.366	2.471	2.566	2.576	2.419	2.265	
	Ebn at Noon	728	852	915	907	914	878	844	856	854	806	722	662	
	Edh at Noon	51	73	92	102	99	109	112	94	75	58	45	40	
All-Sky Solar Radiation	RadAvg	0.78	1.86	3.45	5.00	5.54	5.66	5.47	4.46	3.02	1.58	0.73	0.47	
	RadStd	0.09	0.14	0.27	0.34	0.55	0.48	0.35	0.37	0.35	0.23	0.09	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page